ASignX Data Dictionary

This dictionary documents the current ASignX application schema: 23 tables, 138 fields and 41 foreign keys. It covers account profiles, learning content, enrollment, lesson activity, hearts, XP, recognition and achievements. Supabase manages auth.users separately.

Prepared 8 November 2026 from server/src/db/schema/app.ts and the checked-in migration and service rules. This is a source-based snapshot; the live Supabase deployment was not inspected.

PK means primary key, FK means foreign key and UK means single-column unique. Multiple PK fields form one composite key. Composite foreign keys apply to the combination of fields.

Nullable means the database permits NULL. A dash in Default means no explicit column default. Required values without defaults must be supplied by the caller or a documented trigger. String defaults appear in quotes.

PostgreSQL varchar lengths are character limits. numeric(6,5) means six total digits and five fractional digits. timestamp with time zone represents an instant, displayed using a chosen timezone.

All 23 application tables enable row-level security. Public content reads require publication. Owner-scoped reads apply where defined. Tables without a client policy are accessed through the trusted backend; authorization remains necessary there.

Supabase auth.users is external. Only id (uuid, primary key) is shown in the ERD because it is the application FK target. Auth owns email, verification and credentials; those are not copied into profiles.

PK, FK, unique indexes and checks below come from Drizzle metadata. Trigger and service rules are described separately. FK updates use NO ACTION. CASCADE deletes dependent rows; RESTRICT prevents deleting a referenced parent.

Current persisted lesson recognition uses exercise_attempts and model_versions. Standalone practice storage, saved Live history and automatic achievement awarding are schema capabilities, not verified end-to-end UI workflows. Current Live text history stays in browser memory.

Document structure

One application table follows per page. Each entry lists its fields, exact SQL types, keys, nullability, defaults, constraints and relevant trigger or service rules. Use the separate ERD image to trace relationships.

1 Profiles

Table: profiles

Application profile and authorization role linked to a Supabase identity.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK, FK	No	-	Same UUID as auth.users.id; supplied by the Auth profile-creation trigger, with no column default.
display_name	varchar(100)	-	No	-	Name displayed in the application.
role	varchar(10)	-	No	'learner'	ASignX authorization role: learner or admin. Separate from Supabase Auth database roles.
timezone	varchar(64)	-	No	'UTC'	Valid IANA time zone used when deriving learning streaks.
created_at	timestamptz	-	No	now()	Profile creation time.
updated_at	timestamptz	-	No	now()	Profile modification time; refreshed by the validation trigger on UPDATE.

Keys and database constraints

Primary key: (id).

FK (id) -> auth.users (id); delete CASCADE.

Check: length(btrim(display_name)) > 0.

Check: role in ('learner', 'admin').

Trigger and application rules

Confirmed Auth identities receive learner profiles through database triggers. Profile updates validate timezone and refresh updated_at. Users cannot assign their own admin role.

2 Signs

Table: signs

Reusable sign vocabulary and demonstration media references.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique sign identifier.
gloss	varchar(100)	UK	No	-	Unique vocabulary label, such as HELLO or THANK-YOU.
meaning	text	-	No	-	Meaning of the sign.
category	varchar(60)	-	Yes	-	Simple browsing label, such as Greetings or Numbers; no separate taxonomy is needed.
instructions	text	-	Yes	-	Optional guidance for performing the sign.
demo_path	text	-	Yes	-	Object path for one curated demonstration video in the learning-media Storage bucket.
is_published	boolean	-	No	false	Whether learners may access this sign.
created_at	timestamptz	-	No	now()	Sign creation time.

Keys and database constraints

Primary key: (id).

Unique: (gloss). NULL remains allowed where the field is nullable.

Check: length(btrim(gloss)) > 0.

Check: length(btrim(meaning)) > 0.

Trigger and application rules

Publication and ownership are checked through application services and database rules.

3 Courses

Table: courses

Course metadata, publication state and configurable cover graphics.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique course identifier.
title	varchar(150)	-	No	-	Course title.
description	text	-	Yes	-	Course overview.
cover_size	integer	-	No	100	Main cover text size setting.
cover_angle	integer	-	No	-18	Pattern angle in degrees.
cover_density	integer	-	No	5	Pattern scale and density setting.
cover_weight	varchar(3)	-	No	'600'	Cover text font weight.
cover_font	varchar(12)	-	No	'outfit'	Cover font identifier.
cover_text	varchar(32)	-	No	''	Custom text displayed on the cover.
cover_style	varchar(12)	-	No	'plain'	Cover layout identifier.
cover_tone	varchar(12)	-	No	'sky'	Cover palette identifier.
is_published	boolean	-	No	false	Whether the course is available to learners.
created_at	timestamptz	-	No	now()	Course creation time.

Keys and database constraints

Primary key: (id).

Check: length(btrim(title)) > 0.

Trigger and application rules

Cover options are validated by the admin API. Draft and archived collections are application workflows; there is no separate archived status column.

4 Lessons

Table: lessons

Ordered learning material and configured maximum first-time XP.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique lesson identifier.
course_id	uuid	FK	No	-	References courses.id.
title	varchar(150)	-	No	-	Lesson title.
content	text	-	Yes	-	Instructional text; demonstrations are reached through lesson_signs.
sort_order	integer	-	No	-	Positive lesson position within its course.
xp_reward	integer	-	No	10	Configured maximum first-time lesson XP; allocated among questions by the service.
is_published	boolean	-	No	false	Whether this lesson is visible within its published course.

Keys and database constraints

Primary key: (id).

Unique: (course_id, sort_order).

FK (course_id) -> courses (id); delete RESTRICT.

Check: length(btrim(title)) > 0.

Check: sort_order > 0.

Check: xp_reward >= 0.

Trigger and application rules

Publication and ownership are checked through application services and database rules.

5 Lesson Signs

Table: lesson_signs

Ordered association between lessons and reusable signs.

Field	SQL type	Key	Null	Default	Description
lesson_id	uuid	PK, FK	No	-	References lessons.id; part of the composite primary key.
sign_id	uuid	PK, FK	No	-	References signs.id; part of the composite primary key.
sort_order	integer	-	No	-	Positive demonstration order within the lesson.

Keys and database constraints

Primary key: (lesson_id, sign_id).

Unique: (lesson_id, sort_order).

FK (lesson_id) -> lessons (id); delete CASCADE.

FK (sign_id) -> signs (id); delete RESTRICT.

Check: sort_order > 0.

Trigger and application rules

Publication and ownership are checked through application services and database rules.

Supporting indexes: (sign_id).

6 Exercises

Table: exercises

Ordered choice, true or false, and camera recognition questions.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique question/exercise identifier.
lesson_id	uuid	FK	No	-	References lessons.id.
answer_type	varchar(16)	-	No	'choice'	choice, boolean or recognition; enforced by a check constraint.
prompt	text	-	No	-	Question displayed to the learner.
sign_id	uuid	FK	Yes	-	Optional reference to signs.id for a sign-based question.
sort_order	integer	-	No	-	Positive exercise position within its lesson.

Keys and database constraints

Primary key: (id).

Unique: (lesson_id, sort_order).

FK (lesson_id) -> lessons (id); delete RESTRICT.

FK (sign_id) -> signs (id); delete RESTRICT.

Check: answer_type in ('choice','boolean','recognition').

Check: length(btrim(prompt)) > 0.

Check: sort_order > 0.

Trigger and application rules

Answer types are choice, boolean and recognition. Recognition requires a target sign; published choice questions require valid options. Correctness is withheld from learner question responses.

Supporting indexes: (sign_id).

7 Exercise Options

Table: exercise_options

Options and server-only correctness for choice and true or false questions.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique option identifier.
exercise_id	uuid	FK	No	-	References exercises.id.
option_text	text	-	No	-	Answer choice text displayed to learners.
is_correct	boolean	-	No	false	Marks the correct option; read only by the trusted grading/authoring backend.
sort_order	integer	-	No	-	Positive option order within the exercise.

Keys and database constraints

Primary key: (id).

Unique: (exercise_id, sort_order).

Unique: (exercise_id, id).

Unique index: (exercise_id) WHERE is_correct.

FK (exercise_id) -> exercises (id); delete RESTRICT.

Check: length(btrim(option_text)) > 0.

Check: sort_order > 0.

Trigger and application rules

The partial unique index permits at most one correct option per exercise. Publication validation supplies the additional completeness rules.

8 Enrollments

Table: enrollments

A learner membership in a course.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique enrollment identifier.
user_id	uuid	FK	No	-	References profiles.id.
course_id	uuid	FK	No	-	References courses.id.
enrolled_at	timestamptz	-	No	now()	Enrollment time.

Keys and database constraints

Primary key: (id).

Unique: (user_id, course_id).

FK (user_id) -> profiles (id); delete CASCADE.

FK (course_id) -> courses (id); delete RESTRICT.

Trigger and application rules

The database prevents changing an existing enrollment owner or course.

Supporting indexes: (course_id).

9 Lesson Progress

Table: lesson_progress

Lesson milestones within an enrollment, without a duplicated XP balance.

Field	SQL type	Key	Null	Default	Description
enrollment_id	uuid	PK, FK	No	-	Enrollment identifying the learner and course.
lesson_id	uuid	PK, FK	No	-	References lessons.id; part of the composite primary key.
perfected_at	timestamptz	-	Yes	-	First time all lesson questions have been answered correctly.
started_at	timestamptz	-	No	now()	Time the learner started the lesson.
completed_at	timestamptz	-	Yes	-	Validated completion time; NULL means in progress.

Keys and database constraints

Primary key: (enrollment_id, lesson_id).

FK (enrollment_id) -> enrollments (id); delete CASCADE.

FK (lesson_id) -> lessons (id); delete RESTRICT.

Check: completed_at is null or completed_at >= started_at.

Trigger and application rules

Progress validation keeps lessons within the enrolled course. started_at and completion milestones are persisted; XP is summed from xp_awards.

Supporting indexes: (lesson_id).

10 Exercise Attempts

Table: exercise_attempts

Immutable graded exercise responses, including optional recognition evidence.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique attempt identifier; reuse this ID for a retried submission.
enrollment_id	uuid	FK	No	-	Enrollment identifying the learner and course.
exercise_id	uuid	FK	No	-	References exercises.id.
response_text	text	-	Yes	-	Recognized response label or other stored response evidence.
recognition_confidence	numeric(6, 5)	-	Yes	-	Optional camera recognition score.
model_version_id	uuid	FK	Yes	-	Model version that produced this prediction or response.
selected_option_id	uuid	FK	Yes	-	Required for choice/boolean, NULL for recognition; option must belong to exercise.
is_correct	boolean	-	No	-	Database-trigger grading result, derived from the stored option or trusted model evidence on insert.
attempted_at	timestamptz	-	No	now()	Submission time.

Keys and database constraints

Primary key: (id).

FK (enrollment_id) -> enrollments (id); delete CASCADE.

FK (exercise_id) -> exercises (id); delete RESTRICT.

FK (model_version_id) -> model_versions (id); delete RESTRICT.

FK (exercise_id, selected_option_id) -> exercise_options (exercise_id, id); delete RESTRICT.

Trigger and application rules

The composite option FK ensures an option belongs to its exercise. Database grading and trusted recognition submission validate responses. Attempts cannot be updated.

Supporting indexes: (enrollment_id); (exercise_id, selected_option_id).

11 Learner Hearts

Table: learner_hearts

One shared heart balance per learner and regeneration baseline.

Field	SQL type	Key	Null	Default	Description
user_id	uuid	PK, FK	No	-	Profile owning this record.
hearts	integer	-	No	3	Shared available hearts from zero through three.
regenerated_at	timestamptz	-	No	now()	Baseline used to calculate timed heart regeneration.

Keys and database constraints

Primary key: (user_id).

FK (user_id) -> profiles (id); delete CASCADE.

Check: hearts between 0 and 3.

Trigger and application rules

The service restores one heart every five minutes, capped at three. Camera mismatches do not spend hearts. These timing and grading policies are service rules.

12 Lesson Runs

Table: lesson_runs

Individual lesson attempts, including reduced-XP replays.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique record identifier.
enrollment_id	uuid	FK	No	-	Enrollment identifying the learner and course.
lesson_id	uuid	FK	No	-	Lesson associated with this record.
replay	boolean	-	No	false	Whether this run began as a replay of a perfected lesson.
started_at	timestamptz	-	No	now()	Time this activity started.
completed_at	timestamptz	-	Yes	-	Completion time; NULL while incomplete.
perfect	boolean	-	No	false	Whether all questions in this run were correct.

Keys and database constraints

Primary key: (id).

Unique index: (enrollment_id, lesson_id) WHERE completed_at is null.

FK (enrollment_id) -> enrollments (id); delete CASCADE.

FK (lesson_id) -> lessons (id); delete RESTRICT.

Trigger and application rules

A partial unique index permits one incomplete run per enrollment and lesson. Database validation checks course membership.

13 Lesson Run Questions

Table: lesson_run_questions

Question order, allocated XP and submitted attempt for a particular run.

Field	SQL type	Key	Null	Default	Description
run_id	uuid	PK, FK	No	-	Lesson run associated with this record.
exercise_id	uuid	PK, FK	No	-	Exercise associated with this record.
xp_value	integer	-	No	-	Question XP allocation snapshotted for this run.
sort_order	integer	-	No	-	Snapshotted question position within this run.
attempt_id	uuid	FK	Yes	-	Graded attempt attached after this run question is answered.

Keys and database constraints

Primary key: (run_id, exercise_id).

Unique: (run_id, sort_order).

FK (run_id) -> lesson_runs (id); delete CASCADE.

FK (exercise_id) -> exercises (id); delete RESTRICT.

FK (attempt_id) -> exercise_attempts (id); delete CASCADE.

Check: xp_value>=0.

Trigger and application rules

The composite key identifies a question within a run. Database validation checks that the exercise and attached attempt match the run enrollment and lesson.

14 Xp Awards

Table: xp_awards

Append-only XP ledger used to derive learner and course totals.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique record identifier.
enrollment_id	uuid	FK	No	-	Enrollment identifying the learner and course.
lesson_id	uuid	FK	No	-	Lesson associated with this record.
run_id	uuid	FK	Yes	-	Lesson run associated with this record.
exercise_id	uuid	FK	Yes	-	Exercise associated with this record.
source	varchar(20)	-	No	-	XP origin such as first_question, replay, material or migrated history.
amount	integer	-	No	-	Nonnegative XP amount granted by this ledger entry.
awarded_at	timestamptz	-	No	now()	Time the XP ledger entry was recorded.

Keys and database constraints

Primary key: (id).

Unique index: (enrollment_id, exercise_id) WHERE source in ('claim','first_question').

Unique index: (run_id) WHERE source in ('run_import','replay','material').

FK (enrollment_id) -> enrollments (id); delete CASCADE.

FK (lesson_id) -> lessons (id); delete RESTRICT.

FK (run_id) -> lesson_runs (id); delete CASCADE.

FK (exercise_id) -> exercises (id); delete RESTRICT.

Check: amount>=0.

Check: source in ('legacy','claim','run_import','first_question','replay','material').

Trigger and application rules

Database validation checks enrollment, lesson, run and exercise consistency. Updates are rejected. First-time question claims are unique; replay and material totals are unique per run.

Supporting indexes: (enrollment_id); (run_id).

15 Heart Practice Rounds

Table: heart_practice_rounds

Practice rounds for restoring a heart without awarding XP.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique record identifier.
user_id	uuid	FK	No	-	Profile owning this record.
completed_at	timestamptz	-	Yes	-	Completion time; NULL while incomplete.
created_at	timestamptz	-	No	now()	Record creation time.

Keys and database constraints

Primary key: (id).

Unique index: (user_id) WHERE completed_at is null.

FK (user_id) -> profiles (id); delete CASCADE.

Trigger and application rules

Only one incomplete round is allowed per learner. Three correct answers restore one heart and award no XP.

16 Heart Practice Questions

Table: heart_practice_questions

Three ordered questions assigned to a heart restoration round.

Field	SQL type	Key	Null	Default	Description
round_id	uuid	PK, FK	No	-	Heart restoration round containing this record.
question_index	integer	PK	No	-	Zero-based position in a three-question heart round.
exercise_id	uuid	FK	No	-	Exercise associated with this record.

Keys and database constraints

Primary key: (round_id, question_index).

FK (round_id) -> heart_practice_rounds (id); delete CASCADE.

FK (exercise_id) -> exercises (id); delete RESTRICT.

Check: question_index between 0 and 2.

Trigger and application rules

Positions are 0, 1 and 2. The application chooses previously encountered choice or boolean questions.

17 Heart Practice Submissions

Table: heart_practice_submissions

Graded answer submissions for the questions in a heart restoration round.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	-	Unique record identifier.
round_id	uuid	FK	No	-	Heart restoration round containing this record.
question_index	integer	FK	No	-	Zero-based position in a three-question heart round.
selected_option_id	uuid	FK	No	-	Selected answer option.
is_correct	boolean	-	No	-	Graded correctness of the submitted response.
submitted_at	timestamptz	-	No	now()	Time this practice answer was recorded.

Keys and database constraints

Primary key: (id).

Unique index: (round_id, question_index) WHERE is_correct.

FK (selected_option_id) -> exercise_options (id); delete RESTRICT.

FK (round_id, question_index) -> heart_practice_questions (round_id, question_index); delete CASCADE.

Trigger and application rules

Database grading verifies the selected option against the assigned question. The composite FK identifies that question. At most one correct submission is stored per round question; updates are rejected.

18 Model Versions

Table: model_versions

Model identity, fingerprint and confidence threshold for traceability.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique model version identifier.
version	varchar(50)	UK	No	-	Unique immutable release name.
fingerprint	varchar(128)	UK	Yes	-	Optional unique model artifact fingerprint.
artifact_path	text	-	Yes	-	Optional model artifact location; not required to be a Supabase Storage path.
confidence_threshold	numeric(6, 5)	-	No	0.80000	Acceptance threshold from 0 to 1 for this model version.
created_at	timestamptz	-	No	now()	Model registration time.

Keys and database constraints

Primary key: (id).

Unique: (version). NULL remains allowed where the field is nullable.

Unique: (fingerprint). NULL remains allowed where the field is nullable.

Check: length(btrim(version)) > 0.

Check: artifact_path is null or length(btrim(artifact_path)) > 0.

Check: confidence_threshold between 0 and 1.

Trigger and application rules

Versions and non-null fingerprints are unique. Model records are immutable. Artifact paths are optional; model binaries belong to the recognition deployment, not necessarily Supabase Storage.

19 Practice Attempts

Table: practice_attempts

Schema support for standalone sign practice results.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique practice attempt identifier; reuse on retry.
user_id	uuid	FK	No	-	References profiles.id.
lesson_id	uuid	FK	Yes	-	Optional references lessons.id; NULL means standalone practice or review.
target_sign_id	uuid	FK	No	-	References signs.id; the sign the learner was asked to perform.
predicted_sign_id	uuid	FK	Yes	-	References signs.id; NULL means no supported sign was recognized.
model_version_id	uuid	FK	No	-	Model version that produced this prediction or response.
confidence	numeric(6, 5)	-	No	-	Recognition score between 0 and 1.
is_correct	boolean	-	No	-	Database-trigger result based on target match and the recorded model confidence threshold.
attempted_at	timestamptz	-	No	now()	Practice time.

Keys and database constraints

Primary key: (id).

FK (user_id) -> profiles (id); delete CASCADE.

FK (lesson_id) -> lessons (id); delete RESTRICT.

FK (target_sign_id) -> signs (id); delete RESTRICT.

FK (predicted_sign_id) -> signs (id); delete RESTRICT.

FK (model_version_id) -> model_versions (id); delete RESTRICT.

Check: confidence between 0 and 1.

Trigger and application rules

Database grading validates model confidence and target prediction. Records are immutable. Standalone practice persistence is schema support rather than a verified end-to-end UI workflow.

Supporting indexes: (user_id, attempted_at); (lesson_id); (target_sign_id); (predicted_sign_id); (model_version_id).

20 Recognition Sessions

Table: recognition_sessions

Schema support for saved recognition sessions.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique saved session identifier.
user_id	uuid	FK	No	-	References profiles.id; session owner.
started_at	timestamptz	-	No	now()	Session start time.
ended_at	timestamptz	-	Yes	-	Session end time; NULL while open.
final_text	text	-	Yes	-	Text assembled by the finalization trigger from accepted sign meanings; NULL while open.

Keys and database constraints

Primary key: (id).

FK (user_id) -> profiles (id); delete CASCADE.

Check: ended_at is null or ended_at >= started_at.

Check: final_text is null or ended_at is not null.

Trigger and application rules

Database finalization checks session times and text. Current Live history remains in browser memory; the saved-session workflow is not implemented end to end.

Supporting indexes: (user_id, started_at).

21 Recognition Results

Table: recognition_results

Schema support for ordered predictions within a recognition session.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique prediction-event identifier.
session_id	uuid	FK	No	-	References recognition_sessions.id.
sign_id	uuid	FK	Yes	-	References signs.id; NULL means no supported label was recognized.
model_version_id	uuid	FK	No	-	Model version that produced this prediction or response.
sequence_no	integer	-	No	-	Positive event position within the session.
confidence	numeric(6, 5)	-	No	-	Recognition score between 0 and 1; only accepted labeled results contribute to final_text.
recognized_at	timestamptz	-	No	now()	Prediction time.

Keys and database constraints

Primary key: (id).

Unique: (session_id, sequence_no).

FK (session_id) -> recognition_sessions (id); delete CASCADE.

FK (sign_id) -> signs (id); delete RESTRICT.

FK (model_version_id) -> model_versions (id); delete RESTRICT.

Check: sequence_no > 0.

Check: confidence between 0 and 1.

Trigger and application rules

Database validation checks session ordering and model confidence. Results are immutable. This storage model does not prove a current saved-history UI exists.

Supporting indexes: (sign_id); (model_version_id).

22 Achievements

Table: achievements

Achievement definitions and threshold metrics.

Field	SQL type	Key	Null	Default	Description
id	uuid	PK	No	gen_random_uuid()	Unique achievement identifier.
code	varchar(50)	UK	No	-	Unique stable badge code, such as FIRST_LESSON.
name	varchar(100)	-	No	-	Badge name displayed to learners.
description	text	-	No	-	Description of how to earn the badge.
metric	varchar(25)	-	No	-	One of completed_lessons, successful_practices, total_xp, or streak_days.
threshold	integer	-	No	-	Positive target value for the selected metric.

Keys and database constraints

Primary key: (id).

Unique: (code). NULL remains allowed where the field is nullable.

Check: length(btrim(code)) > 0.

Check: length(btrim(name)) > 0.

Check: length(btrim(description)) > 0.

Check: metric in ('completed_lessons', 'successful_practices', 'total_xp', 'streak_days').

Check: threshold > 0.

Trigger and application rules

Metrics are limited to completed_lessons, successful_practices, total_xp and streak_days. Automatic awarding is not implemented end to end.

23 User Achievements

Table: user_achievements

Association recording an achievement unlocked by a learner.

Field	SQL type	Key	Null	Default	Description
user_id	uuid	PK, FK	No	-	References profiles.id; part of the composite primary key.
achievement_id	uuid	PK, FK	No	-	References achievements.id; part of the composite primary key.
unlocked_at	timestamptz	-	No	now()	Time the backend confirmed eligibility and awarded the badge.

Keys and database constraints

Primary key: (user_id, achievement_id).

FK (user_id) -> profiles (id); delete CASCADE.

FK (achievement_id) -> achievements (id); delete RESTRICT.

Trigger and application rules

Database eligibility validation checks achievement thresholds. Awards are immutable; automatic awarding is not implemented end to end.

Supporting indexes: (achievement_id).